

Global vs Static Variables



global variable also called static variable in java?



No, **global variables** and **static variables** are not the same in Java, but the term **global variable** is often misused to refer to **static variables**. Here's the explanation:

Global Variables

- Java **does not have true global variables** because all variables in Java must belong to a class or method.
- A global variable typically refers to a variable accessible throughout an entire program, irrespective of its location. This concept doesn't exist in Java due to its object-oriented nature.

Static Variables

- Static variables in Java **are often considered an alternative to global variables**.
- They belong to the **class** rather than any specific instance of the class.
- They can be accessed using the class name and are shared among all instances of the class.

Key Points About Static Variables:

- **Declaration:** Declared with the `static` keyword.

```
java

public class Example {
    static int count = 0; // static variable
}
```

- **Shared Across Instances:** A single copy is shared across all objects of the class.
- **Class Scope:** Accessible using the class name without creating an instance.

```
java

System.out.println(Example.count); // Accessed via class name
```

Why Static Variables Are Called Global-Like:

- Because they are accessible from anywhere in the program (provided you have the class reference).
- They do not require an instance of the class to access them.

Example of Static Variable in Action:

java

```
public class GlobalExample {  
    static int sharedValue = 10; // Static variable  
  
    public static void main(String[] args) {  
        System.out.println(GlobalExample.sharedValue); // Access without instance  
    }  
}
```

Conclusion

- Java does not support true global variables due to its object-oriented design.
- **Static variables** are the closest alternative to global variables in Java.